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Tailored scheduling of NBMP media workflows based on the platform capabilities

Abstract

There is included a method and apparatus comprising computer code configured to cause a processor or processors to obtain a request from a network based media processing (NBMP) client to an NBMP workflow manager, transmit a response from the workflow manager to the client, obtain a second request from the client to the workflow manager, control the NBMP workflow manager to set at least one parameter of a NBMP workflow based on the second request, and to control the media content to be processed using the NBMP workflow, the at least one parameter being of any of a scaling scheme and a scheduling scheme, and the requests regard capabilities requested from the workflow manager.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application claims priority to provisional applications U.S. 63/332,608, filed on Apr. 19, 2022, and U.S. 63/332,611, also filed on Apr. 19, 2022, the contents of which are hereby incorporated by reference, in their entirety, into the present application.

BACKGROUND

1. Field

(1) The present disclosure is directed to providing a call flow for scheduling and scaling workflows

using network based media processing (NBMP) descriptors based on the platform capabilities.

2. Description of Related Art

(2) Even if network and cloud platforms are used to run various applications and a network based media processing (NBMP) standard defines a set of tools for the independent processing of media segments thereby, due to technical deficiencies, no call flows are provided in the NBMP specification for any of scheduling and scaling the workflows.

SUMMARY

(3) To address one or more different technical problems, this disclosure provides technical solutions to reduce network overhead and server computational overheads while delivering an option to apply various operations to the resolved element such that in using these operations some of practicality and technical signaling features thereof may be improved.

(4) There is included a method and apparatus comprising memory configured to store computer program code and a processor or processors configured to access the computer program code and operate as instructed by the computer program code. The computer program code comprises first obtaining code configured to cause the at least one processor to obtain a first request from a network based media processing (NBMP) client to an NBMP workflow manager, transmitting code configured to cause the at least one processor to transmit a first response from the NBMP workflow manager to the NBMP client, the first response being in response to and based on the first request, second obtaining code configured to cause the at least one processor to obtain a second request from the NBMP client to the NBMP workflow manager, the second request being in response to and based on the first response, first controlling code configured to cause the at least one processor to control the NBMP workflow manager to set at least one parameter of a NBMP workflow in response to and based on the second request, the at least one parameter being of any of a scaling scheme and a scheduling scheme, and second controlling code configured to cause the at least one processor to control the media content to be processed using the NBMP workflow, the first request and the second request each request capabilities from the NBMP workflow manager, the capabilities being one of scheduling capabilities and scaling capabilities other than of functions of tasks of processing the NBMP workflow.

(5) According to exemplary embodiments, the first request instructs the NBMP workflow manager to respond to the NBMP client with the capabilities from the NBMP workflow manager, the second request instructs the NBMP workflow manager to set the at least one parameter of the NBMP workflow, the parameter being of the capabilities from the NBMP workflow manager and other than of functions of tasks of processing the NBMP workflow, and the NBMP workflow is, at least at a time of the first request, a running workflow

(6) According to exemplary embodiments, the first request requests scheduling capabilities, as the capabilities, of the NBMP workflow manager, and the second request instructs the NBMP workflow manager to set the at least one parameter based on the scheduling capabilities.

(7) According to exemplary embodiments, there is further, of the computer program code, second transmitting code configured to cause the at least one processor to transmit a second response from the NBMP workflow manager to the NBMP client, and the second request instructs the NBMP workflow manager to set a specific schedule for the running workflow, and the second response indicates to the NBMP client whether the specific schedule was set in response to the second request.

(8) According to exemplary embodiments, the at least one parameter defines whether scheduling of the NBMP workflow is based on any of units of duration and units of segments.

(9) According to exemplary embodiments, the first request requests scaling capabilities, as the capabilities, of the NBMP workflow manager, and the second request instructs the NBMP workflow manager to set the at least one parameter based on the scaling capabilities.

(10) According to exemplary embodiments, there is further, of the computer program code, second transmitting code configured to cause the at least one processor to transmit a second response from

the NBMP workflow manager to the NBMP client, and the second request instructs the NBMP workflow manager to set a specific scaling for the running workflow, and the second response indicates to the NBMP client whether the specific scaling was set in response to the second request.

(11) According to exemplary embodiments, the running workflow comprises a workflow descriptor (WD) indicating a workflow descriptor document (WDD), and controlling the NBMP workflow manager to set the at least one parameter of the NBMP workflow in response to and based on the second request comprises adding a descriptor to the WDD during an UpdateWorkflow operation and based on the at least one parameter.

(12) According to exemplary embodiments, the second response comprises returning an updated WDD to the NBMP client based on adding the descriptor to the WDD.

(13) According to exemplary embodiments, the at least one parameter defines scaling by any of upgrading a media processing entity and adding parallel tasks to existing tasks of the running workflow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Further features, nature, and various advantages of the disclosed subject matter will be more apparent from the following detailed description and the accompanying drawings in which:
- (2) FIG. 1 is a simplified schematic illustration of an environment in which methods, apparatuses and systems described herein may be implemented, according to embodiments.
- (3) FIG. 2 is a simplified schematic illustration in accordance with embodiments.
- (4) FIG. 3 is a simplified block diagram regarding decoders in accordance with embodiments.
- (5) FIG. 4 is a simplified block diagram regarding encoders in accordance with embodiments.
- (6) FIG. 5 is a simplified block diagram of a network based media processing (NBMP) system according to embodiments.
- (7) FIG. 6 is a simplified flowchart of a method of processing media content in Moving Picture Experts Group (MPEG) NBMP according to embodiments.
- (8) FIG. 7 is a simplified block diagram of an apparatus for processing media content in MPEG NBMP, according to embodiments.
- (9) FIG. 8 is a simplified block diagram of an updated logical structure of an NBMP system, according to embodiments.
- (10) FIG. 9 is a simplified timing diagram in accordance with embodiments.
- (11) FIG. 10 is a simplified timing diagram in accordance with embodiments.
- (12) FIG. 11 is a simplified block diagram in accordance with embodiments.

DETAILED DESCRIPTION

- (13) The proposed features discussed below may be used separately or combined in any order. Further, the embodiments may be implemented by processing circuitry (e.g., one or more processors or one or more integrated circuits). In one example, the one or more processors execute a program that is stored in a non-transitory computer-readable medium.
- (14) Embodiments described herein provide functional improvements to a Moving Picture Experts Group (MPEG) network based media processing (NBMP) standard. Such improvements increase media processing efficiency, increase speed and lower cost of deployment of media services, and allow large scale deployment of media services by leveraging public, private or hybrid cloud services.
- (15) In examples, the functional improvements to the MPEG NBMP standard include a harmonization of Workflow, Task and Functions, and defining a one-to-one relationship between logical items, data documents and REST resources for each of them. Access to main and basic descriptors may be enabled by creating representational state transfer (REST) resources for every

descriptor. And call flows may be added for any of scheduling and scaling of the workflows.

(16) FIG. 1 is a diagram of an environment **100** in which methods, apparatuses and systems described herein may be implemented, according to embodiments. As shown in FIG. 1, the environment **100** may include a user device **110**, a platform **120**, and a network **130**. Devices of the environment **100** may interconnect via wired connections, wireless connections, or a combination of wired and wireless connections.

(17) The user device **110** includes one or more devices capable of receiving, generating, storing, processing, and/or providing information associated with platform **120**. For example, the user device **110** may include a computing device (e.g., a desktop computer, a laptop computer, a tablet computer, a handheld computer, a smart speaker, a server, etc.), a mobile phone (e.g., a smart phone, a radiotelephone, etc.), a wearable device (e.g., a pair of smart glasses or a smart watch), or a similar device. In some implementations, the user device **110** may receive information from and/or transmit information to the platform **120**.

(18) The platform **120** includes one or more devices as described elsewhere herein. In some implementations, the platform **120** may include a cloud server or a group of cloud servers. In some implementations, the platform **120** may be designed to be modular such that software components may be swapped in or out depending on a particular need. As such, the platform **120** may be easily and/or quickly reconfigured for different uses.

(19) In some implementations, as shown, the platform **120** may be hosted in a cloud computing environment **122**. Notably, while implementations described herein describe the platform **120** as being hosted in the cloud computing environment **122**, in some implementations, the platform **120** may not be cloud-based (i.e., may be implemented outside of a cloud computing environment) or may be partially cloud-based.

(20) The cloud computing environment **122** includes an environment that hosts the platform **120**. The cloud computing environment **122** may provide computation, software, data access, storage, etc. services that do not require end-user (e.g., the user device **110**) knowledge of a physical location and configuration of system(s) and/or device(s) that hosts the platform **120**. As shown, the cloud computing environment **122** may include a group of computing resources **124** (referred to collectively as “computing resources **124**” and individually as “computing resource **124**”).

(21) The computing resource **124** includes one or more personal computers, workstation computers, server devices, or other types of computation and/or communication devices. In some implementations, the computing resource **124** may host the platform **120**. The cloud resources may include compute instances executing in the computing resource **124**, storage devices provided in the computing resource **124**, data transfer devices provided by the computing resource **124**, etc. In some implementations, the computing resource **124** may communicate with other computing resources **124** via wired connections, wireless connections, or a combination of wired and wireless connections.

(22) As further shown in FIG. 1, the computing resource **124** includes a group of cloud resources, such as one or more applications (“APPs”) **124-1**, one or more virtual machines (“VMs”) **124-2**, virtualized storage (“VSs”) **124-3**, one or more hypervisors (“HYPs”) **124-4**, or the like.

(23) The application **124-1** includes one or more software applications that may be provided to or accessed by the user device **110** and/or the platform **120**. The application **124-1** may eliminate a need to install and execute the software applications on the user device **110**. For example, the application **124-1** may include software associated with the platform **120** and/or any other software capable of being provided via the cloud computing environment **122**. In some implementations, one application **124-1** may send/receive information to/from one or more other applications **124-1**, via the virtual machine **124-2**.

(24) The virtual machine **124-2** includes a software implementation of a machine (e.g., a computer) that executes programs like a physical machine. The virtual machine **124-2** may be either a system virtual machine or a process virtual machine, depending upon use and degree of correspondence to

any real machine by the virtual machine **124-2**. A system virtual machine may provide a complete system platform that supports execution of a complete operating system (“OS”). A process virtual machine may execute a single program, and may support a single process. In some implementations, the virtual machine **124-2** may execute on behalf of a user (e.g., the user device **110**), and may manage infrastructure of the cloud computing environment **122**, such as data management, synchronization, or long-duration data transfers.

(25) The virtualized storage **124-3** includes one or more storage systems and/or one or more devices that use virtualization techniques within the storage systems or devices of the computing resource **124**. In some implementations, within the context of a storage system, types of virtualizations may include block virtualization and file virtualization. Block virtualization may refer to abstraction (or separation) of logical storage from physical storage so that the storage system may be accessed without regard to physical storage or heterogeneous structure. The separation may permit administrators of the storage system flexibility in how the administrators manage storage for end users. File virtualization may eliminate dependencies between data accessed at a file level and a location where files are physically stored. This may enable optimization of storage use, server consolidation, and/or performance of non-disruptive file migrations.

(26) The hypervisor **124-4** may provide hardware virtualization techniques that allow multiple operating systems (e.g., “guest operating systems”) to execute concurrently on a host computer, such as the computing resource **124**. The hypervisor **124-4** may present a virtual operating platform to the guest operating systems, and may manage the execution of the guest operating systems. Multiple instances of a variety of operating systems may share virtualized hardware resources.

(27) The network **130** includes one or more wired and/or wireless networks. For example, the network **130** may include a cellular network (e.g., a fifth generation (5G) network, a long-term evolution (LTE) network, a third generation (3G) network, a code division multiple access (CDMA) network, etc.), a public land mobile network (PLMN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a telephone network (e.g., the Public Switched Telephone Network (PSTN)), a private network, an ad hoc network, an intranet, the Internet, a fiber optic-based network, or the like, and/or a combination of these or other types of networks.

(28) The number and arrangement of devices and networks shown in FIG. **1** are provided as an example. In practice, there may be additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of the environment **100** may perform one or more functions described as being performed by another set of devices of the environment **100**.

(29) FIG. **2** illustrates, as an example for an application for the disclosed subject matter, the placement of a video encoder and decoder in a streaming environment. The disclosed subject matter can be equally applicable to other video enabled applications, including, for example, video conferencing, digital TV, storing of compressed video on digital media including CD, DVD, memory stick and the like, and so on.

(30) A streaming system may include a capture subsystem **203**, that can include a video source **201**, for example a digital camera, creating, for example, an uncompressed video sample stream **213**. That sample stream **213** may be emphasized as a high data volume when compared to encoded video bitstreams and can be processed by an encoder **202** coupled to the camera **201**. The encoder **202** can include hardware, software, or a combination thereof to enable or implement aspects of the disclosed subject matter as described in more detail below. The encoded video bitstream **204**, which may be emphasized as a lower data volume when compared to the sample stream, can be

stored on a streaming server **205** for future use. One or more streaming clients **212** and **207** can access the streaming server **205** to retrieve copies **208** and **206** of the encoded video bitstream **204**. A client **212** can include a video decoder **211** which decodes the incoming copy of the encoded video bitstream **208** and creates an outgoing video sample stream **210** that can be rendered on a display **209** or other rendering device (not depicted). In some streaming systems, the video bitstreams **204**, **206** and **208** can be encoded according to certain video coding/compression standards. Examples of those standards are noted above and described further herein.

(31) FIG. 3 may be a functional block diagram of a video decoder **300** according to an embodiment of the present invention.

(32) A receiver **302** may receive one or more codec video sequences to be decoded by the decoder **300**; in the same or another embodiment, one coded video sequence at a time, where the decoding of each coded video sequence is independent from other coded video sequences. The coded video sequence may be received from a channel **301**, which may be a hardware/software link to a storage device which stores the encoded video data. The receiver **302** may receive the encoded video data with other data, for example, coded audio data and/or ancillary data streams, that may be forwarded to their respective using entities (not depicted). The receiver **302** may separate the coded video sequence from the other data. To combat network jitter, a buffer memory **303** may be coupled in between receiver **302** and entropy decoder/parser **304** (“parser” henceforth). When receiver **302** is receiving data from a store/forward device of sufficient bandwidth and controllability, or from an isosynchronous network, the buffer **303** may not be needed, or can be small. For use on best effort packet networks such as the Internet, the buffer **303** may be required, can be comparatively large and can advantageously of adaptive size.

(33) The video decoder **300** may include a parser **304** to reconstruct symbols **313** from the entropy coded video sequence. Categories of those symbols include information used to manage operation of the decoder **300**, and potentially information to control a rendering device such as a display **312** that is not an integral part of the decoder but can be coupled to it. The control information for the rendering device(s) may be in the form of Supplementary Enhancement Information (SEI messages) or Video Usability Information parameter set fragments (not depicted). The parser **304** may parse/entropy-decode the coded video sequence received. The coding of the coded video sequence can be in accordance with a video coding technology or standard, and can follow principles well known to a person skilled in the art, including variable length coding, Huffman coding, arithmetic coding with or without context sensitivity, and so forth. The parser **304** may extract from the coded video sequence, a set of subgroup parameters for at least one of the subgroups of pixels in the video decoder, based upon at least one parameters corresponding to the group. Subgroups can include Groups of Pictures (GOPs), pictures, tiles, slices, macroblocks, Coding Units (CUs), blocks, Transform Units (TUs), Prediction Units (PUs) and so forth. The entropy decoder/parser may also extract from the coded video sequence information such as transform coefficients, quantizer parameter values, motion vectors, and so forth.

(34) The parser **304** may perform entropy decoding/parsing operation on the video sequence received from the buffer **303**, so to create symbols **313**. The parser **304** may receive encoded data, and selectively decode particular symbols **313**. Further, the parser **304** may determine whether the particular symbols **313** are to be provided to a Motion Compensation Prediction unit **306**, a scaler/inverse transform unit **305**, an Intra Prediction Unit **307**, or a loop filter **311**.

(35) Reconstruction of the symbols **313** can involve multiple different units depending on the type of the coded video picture or parts thereof (such as: inter and intra picture, inter and intra block), and other factors. Which units are involved, and how, can be controlled by the subgroup control information that was parsed from the coded video sequence by the parser **304**. The flow of such subgroup control information between the parser **304** and the multiple units below is not depicted for clarity.

(36) Beyond the functional blocks already mentioned, decoder **300** can be conceptually subdivided

into a number of functional units as described below. In a practical implementation operating under commercial constraints, many of these units interact closely with each other and can, at least partly, be integrated into each other. However, for the purpose of describing the disclosed subject matter, the conceptual subdivision into the functional units below is appropriate.

(37) A first unit is the scaler/inverse transform unit **305**. The scaler/inverse transform unit **305** receives quantized transform coefficient as well as control information, including which transform to use, block size, quantization factor, quantization scaling matrices, etc. as symbol(s) **313** from the parser **304**. It can output blocks comprising sample values, that can be input into aggregator **310**.

(38) In some cases, the output samples of the scaler/inverse transform **305** can pertain to an intra coded block; that is: a block that is not using predictive information from previously reconstructed pictures, but can use predictive information from previously reconstructed parts of the current picture. Such predictive information can be provided by an intra picture prediction unit **307**. In some cases, the intra picture prediction unit **307** generates a block of the same size and shape of the block under reconstruction, using surrounding already reconstructed information fetched from the current (partly reconstructed) picture **309**. The aggregator **310**, in some cases, adds, on a per sample basis, the prediction information the intra prediction unit **307** has generated to the output sample information as provided by the scaler/inverse transform unit **305**.

(39) In other cases, the output samples of the scaler/inverse transform unit **305** can pertain to an inter coded, and potentially motion compensated block. In such a case, a Motion Compensation Prediction unit **306** can access reference picture memory **308** to fetch samples used for prediction. After motion compensating the fetched samples in accordance with the symbols **313** pertaining to the block, these samples can be added by the aggregator **310** to the output of the scaler/inverse transform unit (in this case called the residual samples or residual signal) so to generate output sample information. The addresses within the reference picture memory from where the motion compensation unit fetches prediction samples can be controlled by motion vectors, available to the motion compensation unit in the form of symbols **313** that can have, for example X, Y, and reference picture components. Motion compensation also can include interpolation of sample values as fetched from the reference picture memory when sub-sample exact motion vectors are in use, motion vector prediction mechanisms, and so forth.

(40) The output samples of the aggregator **310** can be subject to various loop filtering techniques in the loop filter unit **311**. Video compression technologies can include in-loop filter technologies that are controlled by parameters included in the coded video bitstream and made available to the loop filter unit **311** as symbols **313** from the parser **304**, but can also be responsive to meta-information obtained during the decoding of previous (in decoding order) parts of the coded picture or coded video sequence, as well as responsive to previously reconstructed and loop-filtered sample values.

(41) The output of the loop filter unit **311** can be a sample stream that can be output to the display **312**, which may be a render device, as well as stored in the reference picture memory **557** for use in future inter-picture prediction.

(42) Certain coded pictures, once fully reconstructed, can be used as reference pictures for future prediction. Once a coded picture is fully reconstructed and the coded picture has been identified as a reference picture (by, for example, parser **304**), the current reference picture **309** can become part of the reference picture buffer **308**, and a fresh current picture memory can be reallocated before commencing the reconstruction of the following coded picture.

(43) The video decoder **300** may perform decoding operations according to a predetermined video compression technology that may be documented in a standard, such as ITU-T Rec. H.265. The coded video sequence may conform to a syntax specified by the video compression technology or standard being used, in the sense that it adheres to the syntax of the video compression technology or standard, as specified in the video compression technology document or standard and specifically in the profiles document therein. Also necessary for compliance can be that the complexity of the coded video sequence is within bounds as defined by the level of the video

compression technology or standard. In some cases, levels restrict the maximum picture size, maximum frame rate, maximum reconstruction sample rate (measured in, for example megasamples per second), maximum reference picture size, and so on. Limits set by levels can, in some cases, be further restricted through Hypothetical Reference Decoder (HRD) specifications and metadata for HRD buffer management signaled in the coded video sequence.

(44) In an embodiment, the receiver **302** may receive additional (redundant) data with the encoded video. The additional data may be included as part of the coded video sequence(s). The additional data may be used by the video decoder **300** to properly decode the data and/or to more accurately reconstruct the original video data. Additional data can be in the form of, for example, temporal, spatial, or signal-to-noise ratio (SNR) enhancement layers, redundant slices, redundant pictures, forward error correction codes, and so on.

(45) FIG. **4** may be a functional block diagram of a video encoder **400** according to an embodiment of the present disclosure.

(46) The encoder **400** may receive video samples from a video source **401** (that is not part of the encoder) that may capture video image(s) to be coded by the encoder **400**.

(47) The video source **401** may provide the source video sequence to be coded by the encoder (**303**) in the form of a digital video sample stream that can be of any suitable bit depth (for example: 8 bit, 10 bit, 12 bit, . . .), any colorspace (for example, BT.601 Y CrCb, RGB, . . .) and any suitable sampling structure (for example Y CrCb 4:2:0, Y CrCb 4:4:4). In a media serving system, the video source **401** may be a storage device storing previously prepared video. In a videoconferencing system, the video source **401** may be a camera that captures local image information as a video sequence. Video data may be provided as a plurality of individual pictures that impart motion when viewed in sequence. The pictures themselves may be organized as a spatial array of pixels, wherein each pixel can comprise one or more samples depending on the sampling structure, color space, etc. in use. A person skilled in the art can readily understand the relationship between pixels and samples. The description below focuses on samples.

(48) According to an embodiment, the encoder **400** may code and compress the pictures of the source video sequence into a coded video sequence **410** in real time or under any other time constraints as required by the application. Enforcing appropriate coding speed is one function of Controller **402**. Controller controls other functional units as described below and is functionally coupled to these units. The coupling is not depicted for clarity. Parameters set by controller can include rate control related parameters (picture skip, quantizer, lambda value of rate-distortion optimization techniques, . . .), picture size, group of pictures (GOP) layout, maximum motion vector search range, and so forth. A person skilled in the art can readily identify other functions of controller **402** as they may pertain to video encoder **400** optimized for a certain system design.

(49) Some video encoders operate in what a person skilled in the art readily recognizes as a “coding loop.” As an oversimplified description, a coding loop can consist of the encoding part of an encoder (for example a source coder **403**) (responsible for creating symbols based on an input picture to be coded, and a reference picture(s)), and a (local) decoder **406** embedded in the encoder **400** that reconstructs the symbols to create the sample data that a (remote) decoder also would create (as any compression between symbols and coded video bitstream is lossless in the video compression technologies considered in the disclosed subject matter). That reconstructed sample stream is input to the reference picture memory **405**. As the decoding of a symbol stream leads to bit-exact results independent of decoder location (local or remote), the reference picture buffer content is also bit exact between local encoder and remote encoder. In other words, the prediction part of an encoder “sees” as reference picture samples exactly the same sample values as a decoder would “see” when using prediction during decoding. This fundamental principle of reference picture synchronicity (and resulting drift, if synchronicity cannot be maintained, for example because of channel errors) is well known to a person skilled in the art.

(50) The operation of the “local” decoder **406** can be the same as of a “remote” decoder **300**, which

has already been described in detail above in conjunction with FIG. 3. Briefly referring also to FIG. 4, however, as symbols are available and en/decoding of symbols to a coded video sequence by entropy coder **408** and parser **304** can be lossless, the entropy decoding parts of decoder **300**, including channel **301**, receiver **302**, buffer **303**, and parser **304** may not be fully implemented in local decoder **406**.

(51) An observation that can be made at this point is that any decoder technology except the parsing/entropy decoding that is present in a decoder also necessarily needs to be present, in substantially identical functional form, in a corresponding encoder. The description of encoder technologies can be abbreviated as they are the inverse of the comprehensively described decoder technologies. Only in certain areas a more detail description is required and provided below.

(52) As part of its operation, the source coder **403** may perform motion compensated predictive coding, which codes an input frame predictively with reference to one or more previously-coded frames from the video sequence that were designated as “reference frames.” In this manner, the coding engine **407** codes differences between pixel blocks of an input frame and pixel blocks of reference frame(s) that may be selected as prediction reference(s) to the input frame.

(53) The local video decoder **406** may decode coded video data of frames that may be designated as reference frames, based on symbols created by the source coder **403**. Operations of the coding engine **407** may advantageously be lossy processes. When the coded video data may be decoded at a video decoder (not shown in FIG. 4), the reconstructed video sequence typically may be a replica of the source video sequence with some errors. The local video decoder **406** replicates decoding processes that may be performed by the video decoder on reference frames and may cause reconstructed reference frames to be stored in the reference picture memory **405**, which may be for example a cache. In this manner, the encoder **400** may store copies of reconstructed reference frames locally that have common content as the reconstructed reference frames that will be obtained by a far-end video decoder (absent transmission errors).

(54) The predictor **404** may perform prediction searches for the coding engine **407**. That is, for a new frame to be coded, the predictor **404** may search the reference picture memory **405** for sample data (as candidate reference pixel blocks) or certain metadata such as reference picture motion vectors, block shapes, and so on, that may serve as an appropriate prediction reference for the new pictures. The predictor **404** may operate on a sample block-by-pixel block basis to find appropriate prediction references. In some cases, as determined by search results obtained by the predictor **404**, an input picture may have prediction references drawn from multiple reference pictures stored in the reference picture memory **405**.

(55) The controller **402** may manage coding operations of the video coder **403**, including, for example, setting of parameters and subgroup parameters used for encoding the video data.

(56) Output of all aforementioned functional units may be subjected to entropy coding in the entropy coder **408**. The entropy coder translates the symbols as generated by the various functional units into a coded video sequence, by loss-less compressing the symbols according to technologies known to a person skilled in the art as, for example Huffman coding, variable length coding, arithmetic coding, and so forth.

(57) The transmitter **409** may buffer the coded video sequence(s) as created by the entropy coder **408** to prepare it for transmission via a communication channel **411**, which may be a hardware/software link to a storage device which would store the encoded video data. The transmitter **409** may merge coded video data from the video coder **403** with other data to be transmitted, for example, coded audio data and/or ancillary data streams (sources not shown).

(58) The controller **402** may manage operation of the encoder **400**. During coding, the controller **405** may assign to each coded picture a certain coded picture type, which may affect the coding techniques that may be applied to the respective picture. For example, pictures often may be assigned as one of the following frame types:

(59) An Intra Picture (I picture) may be one that may be coded and decoded without using any

other frame in the sequence as a source of prediction. Some video codecs allow for different types of Intra pictures, including, for example Independent Decoder Refresh Pictures. A person skilled in the art is aware of those variants of I pictures and their respective applications and features.

(60) A Predictive picture (P picture) may be one that may be coded and decoded using intra prediction or inter prediction using at most one motion vector and reference index to predict the sample values of each block.

(61) A Bi-directionally Predictive Picture (B Picture) may be one that may be coded and decoded using intra prediction or inter prediction using at most two motion vectors and reference indices to predict the sample values of each block. Similarly, multiple-predictive pictures can use more than two reference pictures and associated metadata for the reconstruction of a single block.

(62) Source pictures commonly may be subdivided spatially into a plurality of sample blocks (for example, blocks of 4×4, 8×8, 4×8, or 16×16 samples each) and coded on a block-by-block basis. Blocks may be coded predictively with reference to other (already coded) blocks as determined by the coding assignment applied to the blocks' respective pictures. For example, blocks of I pictures may be coded non-predictively or they may be coded predictively with reference to already coded blocks of the same picture (spatial prediction or intra prediction). Pixel blocks of P pictures may be coded non-predictively, via spatial prediction or via temporal prediction with reference to one previously coded reference pictures. Blocks of B pictures may be coded non-predictively, via spatial prediction or via temporal prediction with reference to one or two previously coded reference pictures.

(63) The video coder **400** may perform coding operations according to a predetermined video coding technology or standard, such as ITU-T Rec. H.265. In its operation, the video coder **400** may perform various compression operations, including predictive coding operations that exploit temporal and spatial redundancies in the input video sequence. The coded video data, therefore, may conform to a syntax specified by the video coding technology or standard being used.

(64) In an embodiment, the transmitter **409** may transmit additional data with the encoded video. The source coder **403** may include such data as part of the coded video sequence. Additional data may comprise temporal/spatial/SNR enhancement layers, other forms of redundant data such as redundant pictures and slices, Supplementary Enhancement Information (SEI) messages, Visual Usability Information (VUI) parameter set fragments, and so on.

(65) FIG. 5 is a block diagram of an NBMP system **500**, according to embodiments.

(66) Referring to FIG. 5, the NBMP system **500** includes an NBMP source **510**, an NBMP workflow manager **520**, a function repository **530**, a network controller **540**, one or more media processing entities **550**, a media source **560**, and a media sink **570**.

(67) The NBMP source **510** may receive instructions from a third party entity **580**, may communicate with the NBMP workflow manager **520** via an NBMP workflow API, and may communicate with the function repository **530** via a function discovery API. For example, the NBMP source **510** may send a workflow description document to the NBMP workflow manager **520**, and may read a function description of functions that are stored in a memory of the function repository **530**. The functions may include media processing functions such as, for example, functions of media decoding, feature point extraction, camera parameter extraction, projection method, seam information extraction, blending, post-processing, and encoding. The NBMP source **510** may include at least one processor and a memory that stores code configured to cause the at least processor to perform functions of the NBMP source **510**.

(68) The NBMP source **510** may request the NBMP workflow manager **520** to create workflow including tasks **551** and **552** to be performed by the one or more media processing entities **550**, by sending the workflow description document to the NBMP workflow manager **520**. The workflow description document may include descriptors, each of which may include parameters.

(69) For example, the NBMP source **510** may select one or more of the functions stored in the function repository **530**, and send, to the NBMP workflow manager **520**, the workflow description

document including the descriptors for describing details such as input and output data, the selected one or more of the functions, and requirements for a workflow. The workflow description document may further include a set of task descriptions and a connection map of inputs and outputs of the tasks 551 and 552 to be performed by the one or more of the media processing entities 550. When the NBMP workflow manager 520 receives such information from the NBMP source 510, the NBMP workflow manager 520 may create the workflow by instantiating the tasks 551 and 552 based on function names and connecting the tasks 551 and 552 in accordance with the connection map.

(70) Alternatively or additionally, the NBMP source 510 may request the NBMP workflow manager 520 to create a workflow by using a set of keywords. For example, the NBMP source 510 may send, to the NBMP workflow manager 520, the workflow description document including the set of the keywords that the NBMP workflow manager 520 may use to find appropriate one or more of the functions stored in the function repository 530. When the NBMP workflow manager 520 receives such information from the NBMP source 510, the NBMP workflow manager 520 may create the workflow by searching for the appropriate one or more of the functions, using the keywords that may be specified in a Processing Descriptor of the workflow description document, and by using other descriptors in the workflow description document to provision and connect the tasks 551 and 552.

(71) The NBMP workflow manager 520 may communicate with the function repository 530 via a function discovery API, and may communicate with one or more of the media processing entities 550, through the network controller 540, via an NBMP task API, an NBMP link API, and a function discovery API. The NBMP workflow manager 520 may include at least one processor and a memory that stores code configured to cause the at least processor to perform functions of the NBMP workflow manager 520.

(72) The NBMP workflow manager 520 may use the NBMP task API to setup, configure, manage, and monitor one or more of the tasks 551 and 552 of the workflow that is performable by the one or more media processing entities 550. In embodiments, the NBMP workflow manager 520 may use the NBMP task API to update and destroy the tasks 551 and 552. To configure, manage, and monitor the tasks 551 and 552 of the workflow, the NBMP workflow manager 520 may send messages, such as requests, to one or more of the media processing entities 550, wherein each message may have descriptors, each of which may include parameters. The tasks 551 and 552 may each include one or more media processing functions 554 and one or more configurations 553 for the one or more media processing functions 554.

(73) In embodiments, after receiving the workflow description document from the NBMP source 510 that does not include a list of tasks (e.g., includes a list of keywords instead of a list of tasks), the NBMP workflow manager 520 may select the tasks based on descriptions of the tasks in the workflow description document, to search the function repository 530, via the function discovery API, to find appropriate one or more of the functions to run as the tasks 551 and 552 for the current workflow. For example, the NBMP workflow manager 520 may select the tasks based on keywords that are provided in the workflow description document. After the appropriate one or more of the functions are identified using the keywords or the set of task descriptions that is provided by the NBMP source 510, the NBMP workflow manager 520 may configure the selected tasks in the workflow by using the NBMP task API. For example, the NBMP workflow manager 520 may extract configuration data from information that is received from the NBMP source, and configure the tasks 551 and 552 based on the extracted configuration data.

(74) The one or more media processing entities 550 may be configured to receive media content from the media source 560, process the received media content in accordance with the workflow that includes the tasks 551 and 552 and is created by the NBMP workflow manager 520, and output the processed media content to the media sink 570. The one or more media processing entities 550 may each include at least one processor and a memory that stores code configured to cause the at

least processor to perform functions of the one or more media processing entities **550**.

(75) The network controller **540** may include at least one processor and a memory that stores code configured to cause the at least processor to perform functions of the network controller **540**.

(76) The media source **560** may include memory that stores media and may be integrated with or separate from the NBMP source **510**. In embodiments, the NBMP workflow manager **520** may notify the NBMP source **510** and/or the media source **560** when a workflow is prepared, and the media source **560** may transmit media content to the one or more of the media processing entities **550** based on a notification that the workflow is prepared.

(77) The media sink **570** may include at least one processor and at least one display that is configured to display media content that is processed by the one or more media processing entities **550**.

(78) The third party entity **580** may include at least one processor and a memory that stores code configured to cause the at least processor to perform functions of the third party entity **380**.

(79) As discussed above, messages from the NBMP source **510** (e.g. a workflow description document for requesting creation of a workflow) to the NBMP workflow manager **520**, and messages (e.g. for causing the workflow to be performed) from the NBMP workflow manager **520** to the one or more media processing entities **550** may include descriptors, each of which including parameters. In embodiments, communication between any of components of the NBMP system **500** using an API may include descriptors, each of which including parameters.

(80) FIG. **6** is a flowchart of a method **600** of processing media content in MPEG NBMP, according to embodiments. In some implementations, one or more process blocks of FIG. **6** may be performed by the platform **120** implementing the NBMP system **300**. In some implementations, one or more process blocks of FIG. **6** may be performed by another device or a group of devices separate from or including the platform **120** implementing the NBMP system **300**, such as the user device **110**.

(81) As shown in FIG. **6**, in operation **610**, the method **600** includes obtaining, from an NBMP source, for example NBMP source **310**, a workflow for processing the media content, the workflow having a WD indicating a workflow descriptor document WDD.

(82) In operation **620**, the method **600** includes, based on the workflow, obtaining a task for processing the media content, the task having a TD indicating a TDD.

(83) In operation **630**, the method **600** includes, based on the task, obtaining at least one among the one or more functions from a function repository storing one or more functions for processing the media content, for example function repository **330**, wherein each of the at least one among the one or more functions has an FD indicating an FDD.

(84) In operation **640**, the method **600** includes processing the media content, using the workflow, the task, and the at least one among the one or more functions.

(85) In an embodiment, the workflow may include a workflow representational state transfer (REST) resource (WR), the task may include a task REST resource (TR), and the at least one among the one or more functions may include a function REST resource.

(86) In an embodiment, the WD, the TD, and the FD may be constructed from one or more general descriptors.

(87) In an embodiment, the WDD may include a workflow description object (WO), the TDD may include a task description object (TO), and the FDD may include at least one function description object (FO).

(88) In an embodiment, the WO, the TO, and the at least one FO include at least one JavaScript Object Notation (JSON) object or at least one Extensible Markup Language (XML) element.

(89) In an embodiment, the WDD may include first link object including a first uniform resource locator (URL) indicating a location of the WDD, the TDD may include a second link object including a second uniform resource locator (URL) indicating a location of the TDD, and the FDD may include a third link object including a third uniform resource locator (URL) indicating a

location of the FDD.

(90) In an embodiment, the TD may include a state descriptor indicating a state of the task.

(91) In an embodiment, the state descriptor may indicate that the state of the task is a null state.

(92) In an embodiment, the at least one among the one or more functions may be retrieved from the function repository using a hyper text transfer protocol (HTTP) query including a search key and a search value corresponding to the at least one among the one or more functions.

(93) In an embodiment, the search value may include at least one from an identifier, a name, a description, a brand, or a keyword associated with the at least one among the one or more functions.

(94) Although FIG. 6 shows example blocks of the method **600**, in some implementations, the method **600** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 6. Additionally, or alternatively, two or more of the blocks of the method **600** may be performed in parallel.

(95) FIG. 7 is a diagram of an apparatus **700** for processing media content in MPEG NBMP, according to embodiments. As shown in FIG. 7, the apparatus **700** includes first obtaining code **710**, second obtaining code **720**, third obtaining code **730**, and processing code **740**.

(96) The first obtaining code **710** may be configured to cause the at least one processor to obtain, from an NBMP source such as NBMP source **310**, a workflow for processing the media content, the workflow having a workflow descriptor (WD) indicating a workflow descriptor document (WDD).

(97) The second obtaining code **720** may be configured to cause the at least one processor to obtain, based on the workflow, a task for processing the media content, the task having a task descriptor (TD) indicating a task descriptor document (TDD).

(98) The third obtaining code **730** may be configured to cause the at least one processor to obtain, based on the task, at least one among the one or more functions from a function repository storing one or more functions for processing the media content, for example function repository **330**, wherein each of the at least one among the one or more functions has an FD indicating an FDD.

(99) The processing code **740** may be configured to cause the at least one processor to process the media content, using the workflow, the task, and the at least one among the one or more functions.

(100) In an embodiment, the workflow may include a workflow representational state transfer (REST) resource (WR), the task may include a task REST resource (TR), and the at least one among the one or more functions may include a function REST resource.

(101) In an embodiment, the WD, the TD, and the FD may be constructed from one or more general descriptors.

(102) In an embodiment, the WDD may include a workflow description object (WO), the TDD may include a task description object (TO), and the FDD may include at least one function description object (FO).

(103) In an embodiment, the WO, the TO, and the at least one FO include at least one JavaScript Object Notation (JSON) object or at least one Extensible Markup Language (XML) element.

(104) In an embodiment, the WDD may include first link object including a first uniform resource locator (URL) indicating a location of the WDD, the TDD may include a second link object including a second uniform resource locator (URL) indicating a location of the TDD, and the FDD may include a third link object including a third uniform resource locator (URL) indicating a location of the FDD.

(105) In an embodiment, the TD may include a state descriptor indicating a state of the task.

(106) In an embodiment, the state descriptor may indicate that the state of the task is a null state.

(107) In an embodiment, the at least one among the one or more functions may be retrieved from the function repository using a hyper text transfer protocol (HTTP) query including a search key and a search value corresponding to the at least one among the one or more functions.

(108) In an embodiment, the search value may include at least one from an identifier, a name, a

description, a brand, or a keyword associated with the at least one among the one or more functions.

(109) The current NBMP specification has a concept of template for task and functions, while defining a document for workflow description referred to as a workflow description document (WDD). Furthermore, it is not clear how task and function templates are converted to REST resources. The API documentation does not clearly define the exact syntax of resources, or how what resources are included in acknowledge of a API operation.

(110) An NBMP specification according to embodiments aligns the concepts of logical items, JSON objects/XML documents, and REST resources for the three items: workflow, task and function, and builds a harmonized and aligned structure for all there. Furthermore, it defines the resource format and constraints in acknowledgements, to make the interfaces true REST APIs.

(111) FIG. 8 shows an example 800 of the relationship between the three entities, as logical items, JSON/XML documents and REST resources.

(112) As can be seen in FIG. 8, logical items may be defined as descriptors. The three main descriptors may be a workflow descriptor (WD), a task descriptor (TD) and a function descriptor (FD). WD, TD, and FD may be built as combinations of the basic descriptors, such as a general descriptor, an input descriptor, an output descriptor, and others.

(113) Workflow description objects (WO), task description objects (TO), and function description objects (FO) may be code realizations of the corresponding logic items, for example, WD, TD, and FD, in JSON or XML.

(114) WDD, task description documents (TDD), and function description documents (FDD) may be documents containing WOs, TOs, and FOs. Documents may be objects in JSON and XML document in XML. Note that FDD may differ from WDD and TDD in the sense that it may include one or more FOs.

(115) Workflow resources (WR), task resources (TR), and function resources (FR) may be WDD, TDD, or FDD includes URLs, and therefore may be identified as REST resources.

(116) A major benefit of the example NBMP specification discussed above is that there may be a one to one relationship between main descriptors, documents, and REST resources, which may allow the system to be precisely specified, and may allow interoperable solutions to be built accordingly.

(117) Designating the Main and Basic Descriptors as REST Resources

(118) The current NBMP specification only enables accessing WR, TR and FR. An NBMP specification according to embodiments may make WR, TR, and FR, REST resources, and also enable the main and basic descriptors to become a REST resource. Therefore, these descriptors may be individually accessed using the NBMP APIs. In this design, each main and basic descriptor object of an included WDD, TDD and FDD in any response may include include one “link” object, including a “ref” with value “self” and a URL indicating the location to that object.

(119) Adding the Task Descriptor as a Component of Workflow Descriptor

(120) The current NBMP specification does not include the Task Descriptor as part of workflow. It defines the relationship between them using ConnectionMap descriptor, which also includes the function identifiers.

(121) However, for an NBMP source to have a complete workflow diagram, an NBMP specification according to embodiments may include task descriptors also in the workflow descriptor, as shown for example in Table 1.

(122) TABLE-US-00001 TABLE 1 Adding Task Descriptor to Workflow Descriptor (WD)
Descriptor Description Cardinality Scheme Defines the scheme and specification 1 that this WD is conforming to. General Parameters of the General descriptors 1 are applicable except the following: InputPorts OutputPorts Input Parameters of the Input descriptor 1 Output Parameters of the Output descriptor 1 Task Parameters of the Task Descriptor 0-N Processing Parameters of the Processing descriptor 0-1 are applicable except the following: URL Requirement Parameters of the

Requirement descriptor 1 Client Parameters of the Client Assistance 0-1 Assistance descriptor
Failover Parameters of the Failover descriptor 0-1 Monitoring Parameters of the Monitoring
descriptor 0-1 are applicable except the following: Variable Assertion Parameters of the Assertion
descriptor 0-1 Reporting Parameters of the Reporting descriptor 0-1 Notification Parameters of the
Notification descriptor 0-1

(123) In Table 1, an additional item, task descriptor, is added. With this addition, a workflow
descriptor may describe the complete map and information about a created workflow.

(124) Adding Media Source and Sink to Workflow DAG

(125) The current NBMP specification does not include the media source **560** and sink **570** in its
workflow directed acyclic graph (DAG) description.

(126) An NBMP specification according to embodiments adds these elements to workflow DAG. A
benefit of this approach is that the resource requirements for network connection between media
source **560** and workflow in on hand, and between Workflow and media sink **570** can be described
with the same DAG. This approach simplifies the documentation of requirements as well as the
establishment and management of workflow by NBMP Workflow Manager.

(127) Adding the Task's Life-Cycle State to General Descriptor

(128) The current NBMP specification defines a life-cycle for tasks. The life-cycle has 5 states.
However, the state of the current state of a Task is not described in any descriptors.

(129) An NBMP specification according to embodiments builds based on REST APIs. Each REST
resource must also capture its state. Therefore, the resource maintains the complete status of the
logical item and there is no need to acquire the state from other data structure.

(130) The NBMP specification according to embodiments adds a “state” parameter to general
descriptor, which can be used to describe a task's state. This addition is shown in Table 2.

(131) TABLE-US-00002 TABLE 2 Addition of “state” parameter to General Descriptor Parameter
Data Card- Name Description Type nality Id Provides a unique identification String 1 in the scope
of Repository/ Workflow to the resource. Name Provides a name for identifying String 1 the
resource. Description Provides a human readable String 1 description for the underlying resource.
Brand Provides category information String 1 for the underlying resource Published The date and
time of publication String 1 Time of this document Repository Provides the Repository's URL that
String 1 Location this resource or its corresponding Function Description is located Priority
Provides priority information for Number 0-1 the underlying resource. Execution Provides
execution time of the String 0-1 Time resource InputPorts Includes a map of port information Map
1 where: 1. Map key: Port Identifier 2. Map value: an object with the following parameters: Stream
Id in Input Descriptor Stream Data Type in Input Descriptor OutputPorts Includes a map of port
information Map 1 where: 3. Map key: Port Identifier Map value: an object which is structured:
Stream Id in Output Descriptor Stream Data Type in Output Descriptor State Providing the current
state of the String 1 item in its life-cycle The value of this parameter shall be one of the followings:
Null Instantiated Idle Running In Error Destroyed

(132) As shown in Table 2, and as illustrated in FIG. 3, a new state “Null” may also added to the
life-cycle to capture the initial state of a Task in its life cycle.

(133) Designate NBWP APIs as REST APIs

(134) The current NBMP specification is not clear how the API operations, requests and responses
work. The documentation does not define interoperable APIs at this stage.

(135) An NBMP specification according to embodiments designs NBMP APIs as REST API.
Therefore, all API operations may be implemented as REST methods using HTTP 1. 1, and the
requests and responses may include REST resources, as shown in FIG. 2. Furthermore, the HTTP
status code is used.

(136) Tables 3, 4 and 5 describes the improved NBMP APIs, their requests and responses.

(137) TABLE-US-00003 TABLE 3 Improved Workflow API Operations Request resource
Operation Description requirements Response Requirements CreateWorkflow Create a WR

including If successful, shall include: workflow information needed to 1) HTTP status code 2xx create a workflow 2) Response's body with updated The General descriptor's WR including: Id shall not be included in a) A value for General this request descriptor's Id b) Updated information including endpoint information where to send media data, metadata, and other information If failed, shall include: 1) HTTP status codes 4xx or 5xx 2) Optionally, response's body with updated WR signaling failed descriptors or parameters UpdateWorkflow Update a Updated WR with If successful, shall include: previously identical General's Id, 1) HTTP status code 2xx created previously received in 2) Response's body with updated workflow Create Workflow's WR including: response General descriptor's Id identical to the one in the request Updated information including endpoint information where to send media data, metadata, and other information If failed, shall include: 1) HTTP status codes 4xx or 5xx 2) Response's body with updated WR signaling failed descriptors or parameters DeleteWorkflow Terminate a WR with identical If successful, shall include: HTTP previously General's Id, previously status code 2xx created received in If failed, shall include: workflow CreateWorkflow's 1) HTTP status codes 4xx or 5xx response 2) Response's body with updated WR signaling failed descriptors or parameters RetrieveWorkflow Retrieve a WR with identical If successful, shall include: previously General's Id, previously 1) HTTP status code 2xx configured received in 2) Response's body with updated workflow CreateWorkflow's WR including: response General descriptor's Id identical to the one in the request Updated information including endpoint information where to send media data, metadata, and other information If failed, shall include: 1) HTTP status codes 4xx or 5xx 2) Response's body with updated WR signaling failed descriptors or parameters GetReports Get reports WR with identical If successful, shall include: for a General's Id, previously 1) HTTP status code 2xx previously received in 2) Response's body with updated configured CreateWorkflow's WR including: workflow response and Report General descriptor's Id descriptors identical to the one in the request Updated report descriptors which were included in the request If failed, shall include: 1) HTTP status codes 4xx or 5xx 2) Response's body with updated WR signaling failed descriptors or parameters

(138) TABLE-US-00004 TABLE 4 Improved Task Configuration Operations Request Operation Description Parameters Response Requirements CreateTask Provision to TR including If successful, shall include: run a task information 3) HTTP status code 2xx inside the needed 4) Response's body with updated TR media to create a including: processing workflow a) A value for General descriptor's Id entity The General b) Updated information including descriptor's Id shall endpoint information where to send not be included in media data, metadata, and other this request. information If failed, shall include: 3) HTTP status codes 4xx or 5xx Optionally, response's body with updated TR signaling failed descriptors or parameters UpdateTask Modify the Updated TR with If successful, shall include: task running identical General's 3) HTTP status code 2xx inside the Id, previously 4) Response's body with updated TR media received in including: processing CreateTask's General descriptor's Id identical to entity response the one in the request Updated information including endpoint information where to send media data, metadata, and other information If failed, shall include: 3) HTTP status codes 4xx or 5xx 4) Response's body with updated TR signalling failed descriptors or parameters GetTask Retrieve task TR with identical If successful, shall include: configuration General's Id, 3) HTTP status code 2xx information previously received 4) Response's body with updated TR in CreateTask's including: response General descriptor's Id identical to the one in the request Updated report descriptors which were included in the request If failed, shall include: 3) HTTP status codes 4xx or 5xx Response's body with updated TR signalling failed descriptors or parameters DeleteTask Request to de- TR with identical If successful, shall include: provision the General's Id, 1. HTTP status code 2xx task running previously received If failed, shall include: inside the in CreateTask's HTTP status codes 4xx or 5xx media response. Response's body with updated TR processing signalling failed descriptors or entity parameters

(139) TABLE-US-00005 TABLE 5 Improved Function Discovery API Operations Operation

Description Request Parameters Response DiscoverAll- Discover all Query string shall be empty, If successful, shall include: Functions functions in i.e. only '?' added to the end 1) HTTP status code 2xx the function of Function Repository's URL 2) Response's body including repository exactly one Function Description Document (FDD) containing FDs of all Function in Repository If failed, shall include: 1) HTTP status codes 4xx or 5xx If no Function is found in the repository, the response shall be considered successful and an empty FSD shall be included in the body.

DiscoverSpecific- Discover a Query string shall include the If successful, shall include: Function set of key-value pairs describing the 1) HTTP status code 2xx functions desired properties of the target 2) Response's body including matching a Function. exactly one Function given string Description Document (FDD) of all Function in Repository with the matching values per key If failed, shall include: 2) HTTP status codes 4xx or 5xx If search is performed and no Function is found, the response shall be considered successful and an empty FSD shall be included in the body.

(140) In Tables 3-5, the request and response body (data) may correspond to the REST resources.

(141) Additionally, to make the response a complete REST resource, an NBMP specification according to embodiments may specify that the included WDD in any response may include one "link" object, including a "ref" with value "self" and a URL indicating the location to WDD. In addition, the included TDD in any response may include one "link" object, including a "ref" with value "self" and a URL indicating the location to TDD. Similarly, each included FDD in any response may include one "link" object, including a "ref" with value "self" and a URL indicating the location to FDD. Further, in every included FDD in any response, each Function Descriptor Object (FDO) may include one "link" object, including a "ref" with value "self" and a URL indicating the location of each FDO.

(142) HTTP "Query String"-Based Search for Function Discovery

(143) The current NBMP specification describes discovery operations. But it does not define the format and protocol of the operations. It also specifies that two keys are used for Function search: either by "identifier" or by "name", but not the combination of them.

(144) An NBMP specification according to embodiments improves the discovery operations by using HTTP query string and combination of key-value pairs. So it defines a simple method (HTTP GET) to implement the operations. This allows the combination of keys to be used in the search. In addition, an NBMP specification according to embodiments adds more keys to the search parameters. Therefore, better search can be performed using different aspects of stored Function in the repository.

(145) According to embodiments, a discovery query and a query string may be used to perform a search. The discovery query is to discover one or more Functions in Function Repository by the properties described in the query. The query string may be used to describe these properties; and he query string may include of a set of key-value pairs, separated by a single '&' character. In each key-value pair, the key and value shall be separated by single '=' character. A query string may be added to the end of resource URL after a single '?' character.

(146) Table 6 lists examples of supported keys in the query string.

(147) TABLE-US-00006 TABLE 6 Function query's keys Descriptor and Parameter used for matching Query keys Descriptor Parameter id General Identifier name General Name description General Description brand General Brand keywords Processing Keywords

(148) Further, although a standard NBMP specification may include a schedule descriptor, such standard, due to technical deficiencies, may not define call flows for scheduling a workflow, as such, embodiments described herein disclose a technical solution to such deficiency arising the in computer technology itself.

(149) For example, FIG. 9 shows an example 900 of a timing diagram for one or more variations of such improvements by call flows for scheduling. As shown in FIG. 9, the NBMP client has at least two optional pre-scheduling steps, a combination of S1 and S2 and a combination of S3 and S4. For example, at S1, the NBMP client can request the scheduling capabilities of the platform from

the workflow manager for the running workflow, or at S3, the NBMP client can request the workflow manager to consider a specific scheduling request for the running workflow.

(150) In either case, the workflow manager responds to the NBMP client with its supporting capabilities and/or whether it can accommodate the requested scheduling request. Note that the NBMP client can request option A, S1 and S2, and option B, S3 and S4, in sequence too. The workflow manager collaborates with the platform and tasks to provide the NBMP client the response in either step.

(151) After getting a satisfactory response, the NBMP client can, at S5, request a specific scheduling scheme, with the possibility of tailoring the scheduling scheme to the capabilities and responses that it received from the workflow manager. The workflow manager, at S6, collaborates with the platform and tasks to perform the scheduling and, at S7, send the result back to the NBMP client. The features of that S6 may also include the features shown in the example of FIGS. 6-8 and related descriptions provided herein which are separate to the scheduling features described in the exemplary embodiments of FIG. 9 herein.

(152) According to exemplary embodiments, the following parameters may be used to address the above scheduling schemes: A schedule-type parameter defining whether the scheduling is performed based on units of duration or in units of segments such that with duration, each step of scheduling is defined by a duration of time, and alternatively, with the number of segments, the execution of each task is performed by a fixed number of input segments. For schedule-type 'duration', a schedule table needs to be included according to exemplary embodiments, the rows in the table may show the order of execution, and each row may have the following parameters: List of tasks or task groups to be executed, Start time using cron syntax and semantics, and The duration of time of execution. For schedule-type 'segment', the workflow may be executed task by task, one task at a time for a fixed number of segments that are defined by the parameter "number-of-segments", A flag 'loop' may also be signaled so as to allow for enabling of looping around a defined schedule. A parameter 'status' may include both the type of requests by the NBMP client and the result of the request by the NBMP workflow manager.

(153) Further, although a standard NBMP specification may include a scale descriptor, such standard, due to technical deficiencies, may not define call flows for scaling a workflow, as such, embodiments described herein disclose a technical solution to such deficiency arising in computer technology itself.

(154) For example, FIG. 10 shows an example **1000** of a timing diagram for one or more variations of such improvements by call flows for scaling. As shown in FIG. 10, the NBMP client has at least two optional pre-scaling steps, a combination of S1 and S2 and a combination of S3 and S4. For example, at S1, the NBMP client can request the scaling capabilities of the platform from the workflow manager for the running workflow, or at S3, the NBMP client can request the workflow manager to consider a specific scaling request for the running workflow.

(155) In either case, the workflow manager responds to the NBMP client with its supporting capabilities and/or whether it can accommodate the requested scaling request considering the running workflow. Note that the NBMP client can request option A, S1 and S2, and option B, S3 and S4, in sequence too. The workflow manager collaborates with the platform and tasks to provide the NBMP client the response in either step.

(156) After getting a satisfactory response, the NBMP client can, at S5, request a specific scaling scheme, with the possibility of tailoring the scaling scheme to the capabilities and responses that it received from the workflow manager. The workflow manager, at S6, collaborates with the platform and tasks to perform the scaling and, at S7, send the result back to the NBMP client. The features of that S6 may also include the features shown in the example of FIGS. 6-8 and related descriptions provided herein which are separate to the scaling features described in the exemplary embodiments of FIG. 9 herein. The features of FIG. 9 and FIG. 10 may be implemented any of in sequence and in parallel to each other. That is, scheduling calls and scaling calls may be made between the

NBMP client and workflow manager in sequence, either before or after the other, and in parallel. (157) According to exemplary embodiments, the following parameters may be used to address the above scaling schemes: A “scaling-type” parameter may define a method of scaling to be by either by upgrading one or more media processing entities (MPEs), such as edge servers, cloud resources, and user devices, or by adding parallel tasks to existing tasks using split-merge functions. A “scaling-factor” parameter may define a speed up or a number of parallel paths depending on the type of scaling. A “status” parameter may indicate both the type of requests by the NBMP client and the result of the request by the NBMP workflow manager. Further, the NBMP Client can add a scale descriptor, such as the parameters noted above, to WDD during a UpdateWorkflow operation including the value for the status parameter which defines the type of request. The workflow manager may respond to this request by returning the updated WDD including the updated scale descriptor according to exemplary embodiments.

(158) The techniques described above, can be implemented as computer software using computer-readable instructions and physically stored in one or more computer-readable media or by a specifically configured one or more hardware processors. For example, FIG. 11 shows a computer system 1100 suitable for implementing certain embodiments of the disclosed subject matter.

(159) The computer software can be coded using any suitable machine code or computer language, that may be subject to assembly, compilation, linking, or like mechanisms to create code comprising instructions that can be executed directly, or through interpretation, micro-code execution, and the like, by computer central processing units (CPUs), Graphics Processing Units (GPUs), and the like.

(160) The instructions can be executed on various types of computers or components thereof, including, for example, personal computers, tablet computers, servers, smartphones, gaming devices, internet of things devices, and the like.

(161) The components shown in FIG. 11 for computer system 1100 are exemplary in nature and are not intended to suggest any limitation as to the scope of use or functionality of the computer software implementing embodiments of the present disclosure. Neither should the configuration of components be interpreted as having any dependency or requirement relating to any one or combination of components illustrated in the exemplary embodiment of a computer system 1100.

(162) Computer system 1100 may include certain human interface input devices. Such a human interface input device may be responsive to input by one or more human users through, for example, tactile input (such as: keystrokes, swipes, data glove movements), audio input (such as: voice, clapping), visual input (such as: gestures), olfactory input (not depicted). The human interface devices can also be used to capture certain media not necessarily directly related to conscious input by a human, such as audio (such as: speech, music, ambient sound), images (such as: scanned images, photographic images obtain from a still image camera), video (such as two-dimensional video, three-dimensional video including stereoscopic video).

(163) Input human interface devices may include one or more of (only one of each depicted): keyboard 1101, mouse 1102, trackpad 1103, touch screen 1110, joystick 1105, microphone 1106, scanner 1108, camera 1107.

(164) Computer system 1100 may also include certain human interface output devices. Such human interface output devices may be stimulating the senses of one or more human users through, for example, tactile output, sound, light, and smell/taste. Such human interface output devices may include tactile output devices (for example tactile feedback by the touch-screen 1110, or joystick 1105, but there can also be tactile feedback devices that do not serve as input devices), audio output devices (such as: speakers 1109, headphones (not depicted)), visual output devices (such as screens 1110 to include CRT screens, LCD screens, plasma screens, OLED screens, each with or without touch-screen input capability, each with or without tactile feedback capability-some of which may be capable to output two dimensional visual output or more than three dimensional output through means such as stereographic output; virtual-reality glasses (not depicted), holographic displays and

smoke tanks (not depicted)), and printers (not depicted).

(165) Computer system **1100** can also include human accessible storage devices and their associated media such as optical media including CD/DVD ROM/RW **1120** with CD/DVD **1111** or the like media, thumb-drive **1122**, removable hard drive or solid state drive **1123**, legacy magnetic media such as tape and floppy disc (not depicted), specialized ROM/ASIC/PLD based devices such as security dongles (not depicted), and the like.

(166) Those skilled in the art should also understand that term “computer readable media” as used in connection with the presently disclosed subject matter does not encompass transmission media, carrier waves, or other transitory signals.

(167) Computer system **1100** can also include interface **1199** to one or more communication networks **1198**. Networks **1198** can for example be wireless, wireline, optical. Networks **1198** can further be local, wide-area, metropolitan, vehicular and industrial, real-time, delay-tolerant, and so on. Examples of networks **1198** include local area networks such as Ethernet, wireless LANs, cellular networks to include GSM, 3G, 4G, 5G, LTE and the like, TV wireline or wireless wide area digital networks to include cable TV, satellite TV, and terrestrial broadcast TV, vehicular and industrial to include CANBus, and so forth. Certain networks **1198** commonly require external network interface adapters that attached to certain general-purpose data ports or peripheral buses (**1150** and **1151**) (such as, for example USB ports of the computer system **1100**; others are commonly integrated into the core of the computer system **1100** by attachment to a system bus as described below (for example Ethernet interface into a PC computer system or cellular network interface into a smartphone computer system). Using any of these networks **1198**, computer system **1100** can communicate with other entities. Such communication can be uni-directional, receive only (for example, broadcast TV), uni-directional send-only (for example CANbus to certain CANbus devices), or bi-directional, for example to other computer systems using local or wide area digital networks. Certain protocols and protocol stacks can be used on each of those networks and network interfaces as described above.

(168) Aforementioned human interface devices, human-accessible storage devices, and network interfaces can be attached to a core **1140** of the computer system **1400**.

(169) The core **1140** can include one or more Central Processing Units (CPU) **1141**, Graphics Processing Units (GPU) **1142**, a graphics adapter **1117**, specialized programmable processing units in the form of Field Programmable Gate Areas (FPGA) **1143**, hardware accelerators for certain tasks **1144**, and so forth. These devices, along with Read-only memory (ROM) **1145**, Random-access memory **1146**, internal mass storage such as internal non-user accessible hard drives, SSDs, and the like **1147**, may be connected through a system bus **1148**. In some computer systems, the system bus **1148** can be accessible in the form of one or more physical plugs to enable extensions by additional CPUs, GPU, and the like. The peripheral devices can be attached either directly to the core's system bus **1148**, or through a peripheral bus **1151**. Architectures for a peripheral bus include PCI, USB, and the like.

(170) CPUs **1141**, GPUs **1142**, FPGAs **1143**, and accelerators **1144** can execute certain instructions that, in combination, can make up the aforementioned computer code. That computer code can be stored in ROM **1145** or RAM **1146**. Transitional data can be also be stored in RAM **1146**, whereas permanent data can be stored for example, in the internal mass storage **1147**. Fast storage and retrieval to any of the memory devices can be enabled through the use of cache memory, that can be closely associated with one or more CPU **1141**, GPU **1142**, mass storage **1147**, ROM **1145**, RAM **1146**, and the like.

(171) The computer readable media can have computer code thereon for performing various computer-implemented operations. The media and computer code can be those specially designed and constructed for the purposes of the present disclosure, or they can be of the kind well known and available to those having skill in the computer software arts.

(172) As an example and not by way of limitation, an architecture corresponding to computer

system **1100**, and specifically the core **1140** can provide functionality as a result of processor(s) (including CPUs, GPUs, FPGA, accelerators, and the like) executing software embodied in one or more tangible, computer-readable media. Such computer-readable media can be media associated with user-accessible mass storage as introduced above, as well as certain storage of the core **1140** that are of non-transitory nature, such as core-internal mass storage **1147** or ROM **1145**. The software implementing various embodiments of the present disclosure can be stored in such devices and executed by core **1140**. A computer-readable medium can include one or more memory devices or chips, according to particular needs. The software can cause the core **1140** and specifically the processors therein (including CPU, GPU, FPGA, and the like) to execute particular processes or particular parts of particular processes described herein, including defining data structures stored in RAM **1146** and modifying such data structures according to the processes defined by the software. In addition or as an alternative, the computer system can provide functionality as a result of logic hardwired or otherwise embodied in a circuit (for example: accelerator **1144**), which can operate in place of or together with software to execute particular processes or particular parts of particular processes described herein. Reference to software can encompass logic, and vice versa, where appropriate. Reference to a computer-readable media can encompass a circuit (such as an integrated circuit (IC)) storing software for execution, a circuit embodying logic for execution, or both, where appropriate. The present disclosure encompasses any suitable combination of hardware and software.

(173) While this disclosure has described several exemplary embodiments, there are alterations, permutations, and various substitute equivalents, which fall within the scope of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise numerous systems and methods which, although not explicitly shown or described herein, embody the principles of the disclosure and are thus within the spirit and scope thereof.

Claims

1. A method of processing a media content in Moving Picture Experts Group (MPEG) Network Based Media Processing (NBMP), the method being performed by at least one processor, and the method comprising: obtaining a first request from an NBMP client to an NBMP workflow manager; transmitting a first response from the NBMP workflow manager to the NBMP client, the first response being in response to and based on the first request; obtaining a second request from the NBMP client to the NBMP workflow manager, the second request being in response to and based on the first response; controlling the NBMP workflow manager to set a plurality of parameters of a NBMP workflow in response to and based on the second request, the parameters comprising a parameter “loop” enabling looping around a defined schedule, a parameter “status”, indicating types of the first request, the first response, and the second request, and at least one of a parameter “duration”, representing a schedule table comprising rows in order of execution of tasks or task groups, and a “segment” parameter indicating task execution task by task for a fixed number of segments; and controlling a media content to be processed using the NBMP workflow and at least one parameter of the plurality of parameters, wherein the first request and the second request each request capabilities from the NBMP workflow manager, the capabilities being scheduling capabilities other than of functions of tasks of processing the NBMP workflow, and wherein the NBMP workflow is, when the first request is obtained, a running workflow.
2. The method according to claim 1, wherein the first request instructs the NBMP workflow manager to respond to the NBMP client with the capabilities from the NBMP workflow manager, the second request instructs the NBMP workflow manager to set the at least one parameter of the NBMP workflow, the at least one parameter being of the capabilities from the NBMP workflow manager and other than of functions of tasks of processing the NBMP workflow.
3. The method according to claim 2, further comprising, before obtaining the first request and

transmitting the first response, obtaining, by the NBMP workflow manager, an initial request from the NBMP client and transmitting, by the NBMP workflow manager, an initial response to the NBMP client, wherein the initial request requests different scheduling capabilities than does the first request, wherein the first request is transmitted from the NBMP client to the NBMP workflow manager in response to the NBMP client receiving the initial response, wherein the first request requests the scheduling capabilities, as the capabilities, of the NBMP workflow manager, and wherein the second request instructs the NBMP workflow manager to set the at least one parameter based on the scheduling capabilities.

4. The method according to claim 3, further comprising: transmitting a second response from the NBMP workflow manager to the NBMP client, wherein the second request instructs the NBMP workflow manager to set a specific schedule for the running workflow, and the second response indicates to the NBMP client whether the specific schedule was set in response to the second request.

5. The method according to claim 2, wherein the at least one parameter defines whether scheduling of the NBMP workflow is based on any of units of duration and units of segments.

6. The method according to claim 2, further comprising obtaining a third request from the NBMP client to the NBMP workflow manager, the third request requests scaling capabilities of the NBMP workflow manager; and obtaining a fourth request from the NBMP client to the NBMP workflow manager, the fourth request instructs the NBMP workflow manager to set a scaling parameter based on the scaling capabilities.

7. The method according to claim 6, further comprising: transmitting a third response from the NBMP workflow manager to the NBMP client, wherein the third response indicates to the NBMP client whether a specific scaling was set in response to the fourth request.

8. The method according to claim 7, wherein the running workflow comprises a workflow descriptor (WD) indicating a workflow descriptor document (WDD), and controlling the NBMP workflow manager to set the at least one parameter of the NBMP workflow in response to and based on the second request comprises adding a descriptor to the WDD during an UpdateWorkflow operation and based on the at least one parameter.

9. The method according to claim 8, wherein the second response comprises returning an updated WDD to the NBMP client based on adding the descriptor to the WDD.

10. The method according to claim 2, wherein the at least one parameter defines scaling by any of upgrading a media processing entity and adding parallel tasks to existing tasks of the running workflow.

11. An apparatus for processing a media content in Moving Picture Experts Group (MPEG) Network Based Media Processing (NBMP), the apparatus comprising: at least one memory configured to store computer program code; at least one processor configured to access the computer program code and operate as instructed by the computer program code, the computer program code including: first obtaining code configured to cause the at least one processor to obtain a first request from an NBMP client to an NBMP workflow manager; transmitting code configured to cause the at least one processor to transmit a first response from the NBMP workflow manager to the NBMP client, the first response being in response to and based on the first request; second obtaining code configured to cause the at least one processor to obtain a second request from the NBMP client to the NBMP workflow manager, the second request being in response to and based on the first response; first controlling code configured to cause the at least one processor to control the NBMP workflow manager to a plurality of parameters of a NBMP workflow in response to and based on the second request, the parameters comprising a parameter "loop" enabling looping around a defined schedule, a parameter "status", indicating types of the first request, the first response, and the second request, and at least one of a parameter "duration", representing a schedule table comprising rows in order of execution of tasks or task groups, and a "segment" parameter indicating task execution task by task for a fixed number of segments; and

second controlling code configured to cause the at least one processor to control a media content to be processed using the NBMP workflow and at least one parameter of the plurality of parameters, wherein the first request and the second request each request capabilities from the NBMP workflow manager, the capabilities being scheduling capabilities other than of functions of tasks of processing the NBMP workflow, and wherein the NBMP workflow is, when the first request is obtained, a running workflow.

12. The apparatus according to claim 11, wherein the first request instructs the NBMP workflow manager to respond to the NBMP client with the capabilities from the NBMP workflow manager, and the second request instructs the NBMP workflow manager to set the at least one parameter of the NBMP workflow, the at least one parameter being of the capabilities from the NBMP workflow manager and other than of functions of tasks of processing the NBMP workflow.

13. The apparatus according to claim 12, wherein the first request requests the scheduling capabilities, as the capabilities, of the NBMP workflow manager, and the second request instructs the NBMP workflow manager to set the at least one parameter based on the scheduling capabilities.

14. The apparatus according to claim 13, further comprising: second transmitting code configured to cause the at least one processor to transmit a second response from the NBMP workflow manager to the NBMP client, wherein the second request instructs the NBMP workflow manager to set a specific schedule for the running workflow, and the second response indicates to the NBMP client whether the specific schedule was set in response to the second request.

15. The apparatus according to claim 12, wherein the at least one parameter defines whether scheduling of the NBMP workflow is based on any of units of duration and units of segments.

16. The apparatus according to claim 12, wherein the NBMP workflow manager is configured to receive and respond to requests for scaling capabilities from the NBMP client.

17. The apparatus according to claim 16, further comprising: second transmitting code configured to cause the at least one processor to transmit a second response from the NBMP workflow manager to the NBMP client, wherein the second response indicates to the NBMP client whether a specific scaling was set.

18. The apparatus according to claim 17, wherein the running workflow comprises a workflow descriptor (WD) indicating a workflow descriptor document (WDD), and controlling the NBMP workflow manager to set the at least one parameter of the NBMP workflow in response to and based on the second request comprises adding a descriptor to the WDD during an UpdateWorkflow operation and based on the at least one parameter.

19. The apparatus according to claim 18, wherein the second response comprises returning an updated WDD to the NBMP client based on adding the descriptor to the WDD.

20. A non-transitory computer readable medium storing a program causing a computer to execute a process for processing a media content in Moving Picture Experts Group (MPEG) Network Based Media Processing (NBMP), the method being performed by at least one processor, and the method comprising, the process comprising: obtaining a first request from an NBMP client to an NBMP workflow manager; transmitting a first response from the NBMP workflow manager to the NBMP client, the first response being in response to and based on the first request; obtaining a second request from the NBMP client to the NBMP workflow manager, the second request being in response to and based on the first response; controlling the NBMP workflow manager to set a plurality of parameters of a NBMP workflow in response to and based on the second request, the parameters comprising a parameter "loop" enabling looping around a defined schedule, a parameter "status", indicating types of the first request, the first response, and the second request, and at least one of a parameter "duration", representing a schedule table comprising rows in order of execution of tasks or task groups, and a "segment" parameter indicating task execution task by task for a fixed number of segments; and controlling a media content to be processed using the NBMP workflow, wherein the first request and the second request each request capabilities from the NBMP workflow manager, the capabilities being scheduling capabilities other than of functions of

tasks of processing the NBMP workflow, and wherein the NBMP workflow is, at least at a time of the first request, a running workflow.
